

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, Computer Architecture, System Design, Data Structures

Experience

• New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

ML Software Engineer

New York, US

- Designed and optimized **computer vision** pipelines (**YOLOv8**, **EfficientNet**) for food-image classification with **image preprocessing**, **image augmentation**, and cloud inference deployment on production endpoints.
- Trained **deep learning** models including **LSTM** time-series detectors on wearable telemetry; executed experiments with **hyperparameter tuning** and monitored outputs through **Python** ETL into **PostgreSQL** for **250K+** records.
- Integrated ML models into a mobile health platform processing real-time sensor streams; collaborated with cross-functional partners to refine model behavior from field feedback.

• Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Built an internal console for image and video evidence review across fulfillment centers; supported backend **Java** services and **AWS** deployment pipelines with automated testing before production release.
- Developed **ML infrastructure** on **ECS Fargate**, **SQS**, and **SNS** with **AWS CDK**; analyzed test failures and incorporated review feedback to improve operational reliability.

• PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bengaluru, India

- Deployed an **NLP/vision OCR neural model** on **Azure** for document image extraction; reduced processing time by **75%** through model-driven automation and parallel batch ingestion.
- Developed production **Python/Java** services supporting analytics integrations; led code reviews and mentored engineers on testing discipline in a small delivery team.

• CampusMesh

Jan 2026 – Present

Software Engineer

New York, US

- Built **embedding-based semantic search** pipelines for an AI assistant; maintained **Python/Node.js** services and production workflows across **4 AWS environments**.

Projects

• Stroke Segmentation & Classification (Deep Learning Research) | Python, TensorFlow, Keras, OpenCV, Scikit-image, NumPy, Deep Learning

- Developed a supervised **deep learning** system for medical **image segmentation** and classification; applied **image preprocessing** and **machine learning model training** workflows published in **IEEE**.

• High-Efficiency ResNet Image Classifier | Python, TensorFlow, Keras, Image Augmentation, Hyperparameter Tuning, CIFAR-10

- Designed a **ResNet** architecture under a strict parameter budget; applied **image augmentation** and **hyperparameter tuning** (Lookahead optimizer) to improve classification accuracy on **CIFAR-10**.

• Hospital Management System (Vision-Enabled Diagnostics) | Python, Hugging Face, AWS SageMaker, Computer Vision, Deep Learning

- Fine-tuned a transformer-based diagnostic model reaching **88% accuracy**; deployed real-time inference on **AWS SageMaker** with preprocessing pipelines for clinical image inputs.

Skills

- **Programming Languages:** Python, C++, Java, JavaScript, SQL
- **Computer Vision & ML:** Computer Vision, Deep Learning, Image Segmentation, Image Preprocessing, Image Augmentation, Machine Learning Model Training, Hyperparameter Tuning
- **Frameworks & Libraries:** TensorFlow, Keras, OpenCV, Scikit-image, NumPy, PyTorch, Hugging Face, scikit-learn
- **Cloud & Deployment:** AWS (SageMaker, Lambda, ECS, S3, CloudWatch), Docker, CI/CD, Git, Linux
- **Databases:** PostgreSQL, MySQL, DynamoDB, MongoDB

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals